📋 Internship Project — Week 2 Project Title: Customer Churn Prediction & Risk Segmentation Dashboard

Assigned Date: 04/05/2026
Level: Mid–Advanced

Submission Date: 10/05/2026
Domain: AI & Data Analytics

🎯 PROBLEM STATEMENT

Subscription-based businesses lose significant revenue every year due to customer churn — customers who cancel or stop using a service. Predicting which customers are likely to churn before they actually leave allows the business to act proactively: offer discounts, improve support, or tailor communication.

Your task is to build an end-to-end Customer Churn Prediction pipeline using a real-world telecom dataset. You will perform exploratory data analysis (EDA), engineer meaningful features, train and compare multiple classification models, evaluate them properly, and — the advanced part — segment customers into risk tiers (High / Medium / Low churn risk) and visualize the results in a clean dashboard using Plotly or Streamlit.

This project simulates a real analyst task at a product or growth team. Focus on interpretability — a business stakeholder should understand your findings, not just your code.

📦 DATASET — WHERE TO GET IT

Go to this link and download the dataset: 🔗
https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Steps to download:

1. Create or log in to your free Kaggle account at www.kaggle.com
2. Search for "Telco Customer Churn" by BlastChar
3. Download the file — WA_Fn-UseC_-Telco-Customer-Churn.csv (~977 KB, 7,043 rows)
4. Use the full dataset — no row limit needed for this project

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✅ TASKS TO COMPLETE

Complete all tasks inside a Jupyter Notebook (.ipynb). Each task must be clearly labeled as a Markdown heading.

TASK 1 — Data Loading & Exploratory Analysis

→ Load the CSV using Pandas and display shape, column types, and first 10 rows

→ Identify the target column (Churn — Yes/No) and check class imbalance

→ Find missing/null values and decide how to handle them (drop vs. impute)

→ Compute summary statistics for numerical columns (mean, median, std)

→ Plot a correlation heatmap of numerical features using Seaborn

TASK 2 — Data Preprocessing & Feature Engineering

→ Convert TotalCharges to numeric — handle coercion errors carefully

→ Encode all categorical columns using Label Encoding or One-Hot Encoding — justify your choice

→ Create at least 2 new features, for example: - ChargesPerMonth = TotalCharges / tenure - SeniorWithNoSupport = (SeniorCitizen == 1) & (TechSupport == "No")

→ Scale numerical features using StandardScaler before model training

→ Split data into 80% train / 20% test using stratified split to preserve class ratio

TASK 3 — Model Training & Comparison

→ Train 3 classification models: - Model 1: Logistic Regression - Model 2: Random Forest Classifier - Model 3: XGBoost (or Gradient Boosting Classifier)

→ Evaluate each model with: Accuracy, Precision, Recall, F1-Score, ROC-AUC

→ Plot a Confusion Matrix for each model side-by-side

→ Plot the ROC Curve for all 3 models on one chart for comparison

→ Perform basic hyperparameter tuning on your best model using GridSearchCV or RandomizedSearchCV [ADVANCED STEP]

TASK 4 — Customer Risk Segmentation [ADVANCED]

→ Use the best model's predicted probability of churn (not just 0/1 prediction)

→ Segment customers into 3 risk tiers based on churn probability:

- 🔴 High Risk → probability ≥ 0.70

- 🟡 Medium Risk → probability between 0.40 and 0.69

- 🟢 Low Risk → probability < 0.40

→ Count how many customers fall into each tier

→ For each tier, compute average MonthlyCharges, average tenure, and Contract type distribution

→ Create a grouped bar chart comparing the 3 tiers across these key features

TASK 5 — Visualizations (Minimum 4 Charts)

→ Chart 1: Feature importance bar chart (top 10 features) from Random Forest or XGBoost
→ Chart 2: Churn rate by Contract Type (Month-to-Month vs. One Year vs. Two Year)

→ Chart 3: Distribution of tenure — churned vs. not-churned customers (overlapping histogram or KDE plot)

→ Chart 4: Risk tier donut/pie chart showing customer count per tier

→ Bonus Chart (Optional +Points): Interactive scatter plot using Plotly — MonthlyCharges vs. Tenure, colored by churn label

TASK 6 — Insights & Business Recommendations

Write a structured section (8–12 lines) inside your notebook answering all of the following:

→ Which model performed best and why did you select it?

→ What are the top 3 factors driving customer churn according to your model?

→ What do High Risk customers have in common? (contract type, charges, tenure patterns)

→ Write 2 specific, actionable business recommendations the company should implement

→ What are the limitations of your model? What could be improved with more data or time?

🛠️ TOOLS & LIBRARIES TO USE

Tool	Purpose
Python 3.x	**Main programming language**
Jupyter / Colab Notebook	**Write and run your code**
Pandas	**Data loading and manipulation**
NumPy	**Numerical operations**
Scikit-learn	**Models, metrics, preprocessing**

XGBoost	**Gradient boosting model**
Matplotlib / Seaborn	**Static charts and heatmaps**
Plotly	**Interactive visualizations**

📁 WHAT TO SUBMIT

Create a folder named: ChurnAnalysis_[YourName]

Inside the folder, include all of the following:

✅ analysis.ipynb — Complete Jupyter Notebook with all 6 tasks clearly labeled

✅ WA_Fn-UseC_-Telco-Customer-Churn.csv — Original dataset (unmodified)

✅ model_comparison.png — ROC curve chart comparing all 3 models

✅ charts/ — A folder containing all 4+ visualization images saved as .png

✅ summary.pdf or summary.docx — 1 to 2 page business report with key findings and recommendations

✅ requirements.txt — List of all Python libraries used (Generate it by running: pip freeze > requirements.txt) [ADVANCED STEP]

Submit the entire folder as a ZIP file via: →

https://docs.google.com/forms/d/e/1FAIpQLSeNV5yk4IUbzEAmVAZo44Q8qXQgDFEIsWtxjS61PraTDMTueQ/viewform?usp=sharing&ouid=114324793138831463483